

Comprehensive Reproduction Report for the Nonsymmorphic Discrete Time Crystal Model

Reproduction notes, numerical implementation, and finite-size checks¹

¹*Local CUDA and H100 reproduction workflow*

(Dated: June 4, 2026)

Abstract

This report summarizes the reproduction of the nonsymmorphic discrete time crystal model of Hu, Fu, Li, and Shen, Phys. Rev. Research 5, L032024 (2023). The work includes a reading-stage review of the mechanism, a noninteracting two-level reproduction with fourth-order Magnus and Runge–Kutta checks, and an interacting exact state-vector spin-chain reproduction with local and H100-ready workflows. The noninteracting reproduction confirms the special refined point $\alpha_{13} = 80.04624211689759$, where one-period return to the initial instantaneous eigenstate is suppressed and a robust two-period response appears. For the interacting chain, the final convention keeps the paper-label interaction strength $J/\hbar\omega = 0.2$ but applies a Pauli-operator scale factor of $1/2$, so $J_{\text{eff}} = 0.1$ in the implemented $\sigma_x^i \sigma_x^{i+1}$ term. Under this convention the refined α_{13} data show the expected lifetime enhancement from $L = 8$ to $L = 10$. The remaining unresolved discrepancy is that the detuned $\alpha = 81.60$ raw $Z(n)$ trace decays much faster than the paper’s Fig. 4(c) under both open and periodic boundary conditions, indicating that the paper’s lifetime extraction, interaction normalization, or numerical protocol may differ from the current raw-observable implementation.

I. SCOPE AND DELIVERABLES

The target paper proposes a solvable route to discrete time-crystal (DTC) dynamics enforced by a nonsymmorphic dynamical symmetry. The reproduction was organized into four stages:

1. read and summarize the main paper and supplemental material;
2. reproduce the noninteracting two-level dynamics and compare fourth-order Magnus evolution with a Runge–Kutta benchmark;
3. reproduce the interacting spin-chain response with CUDA/PyTorch exact state-vector dynamics and HPC-ready parameter files;
4. collect the theory, numerical methods, figures, diagnostics, and open issues into the present comprehensive report.

The main project artifacts are the reading report, the noninteracting reproduction report, the interacting reproduction report, the Python scripts, parameter JSON files, selected

data tables, and the final report in this directory. The numerical code follows a simple function-based structure so that the physical assumptions are visible at the point of use.

II. MODEL AND DTC MECHANISM

The single-spin Hamiltonian used throughout the reproduction is

$$H_0(t) = \frac{1}{2}\Omega \sin(\omega t)\sigma_x + \Omega \sin^2\left(\frac{\omega t}{2}\right)\sigma_y, \quad (1)$$

with the dimensionless convention

$$\hbar = 1, \quad \omega = 1, \quad T = 2\pi, \quad \Omega = \alpha/8. \quad (2)$$

The paper's special parameters α_n are selected by the nonsymmorphic dynamical symmetry. At these points the instantaneous eigenstates swap after one driving period and return only after two periods, producing a subharmonic response at half the drive frequency.

In the interacting stage the implemented chain Hamiltonian is

$$H(t) = \sum_{i=1}^L H_{0,i}(t) + J_{\text{eff}} \sum_{\langle i,j \rangle} \sigma_x^i \sigma_x^j. \quad (3)$$

The final production convention is

$$J_{\text{paper}}/\hbar\omega = 0.2, \quad J_{\text{eff}} = 0.5J_{\text{paper}} = 0.1. \quad (4)$$

This distinction is important because the code multiplies explicit Pauli matrices, whereas the paper's wording uses spin operators. Directly using $J = 0.2$ as the Pauli-product coefficient was found to overdrive the many-body chain and to produce an unphysical early decay for $L = 10$ at refined α_{13} .

The initial state for the interacting runs is

$$|\psi(0)\rangle = \bigotimes_{i=1}^L |+\rangle_i, \quad (5)$$

matching the main text statement that the spins are initially polarized in the x direction. The monitored observables are

$$m_x(t) = \frac{1}{L} \sum_i \langle \psi(t) | \sigma_x^i | \psi(t) \rangle \quad (6)$$

and the stroboscopic signal

$$Z(n) = (-1)^n m_x(nT). \quad (7)$$

The supplemental material also allows more general x -product states $\otimes_i |s_i, x\rangle$, but for global m_x such states can change the amplitude by partial cancellation. The all- $+x$ state therefore gives the cleanest direct comparison with Fig. 4.

III. NONINTERACTING REPRODUCTION

A. Numerical method

The noninteracting solver uses two independent time-evolution checks. The first is a fourth-order Gauss–Legendre Magnus integrator, chosen because it preserves the unitary structure more faithfully than a generic explicit integrator. The second is a classical fourth-order Runge–Kutta solver used as an independent benchmark. Both solvers evolve the same two-level Hamiltonian in Eq. (1).

The refined special point used in all later runs is

$$\alpha_{13} = 80.04624211689759, \quad (8)$$

which is close to the paper’s tabulated value 80.07. The computed one-period return probability at this refined point is 3.57×10^{-18} , consistent with nearly perfect swapping rather than one-period return.

B. Results

Figure 1 compares $\langle \sigma_x(t) \rangle$ for the refined α_{13} and detuned $\alpha = 81.60$. The refined parameter produces a clean half-frequency response over the plotted time window. The detuned parameter still has visible subharmonic character over short times, but the detailed waveform and spectrum shift away from the exact- α_n behavior.

The fractional-period projection data in Fig. 2 sample $0, T/4, T/2, 3T/4$ within each period. This was added because stroboscopic-only plots can hide the intra-period exchange of instantaneous eigenstate weight.

The refined point also shows the expected two-period stroboscopic structure, as summarized in Fig. 3.

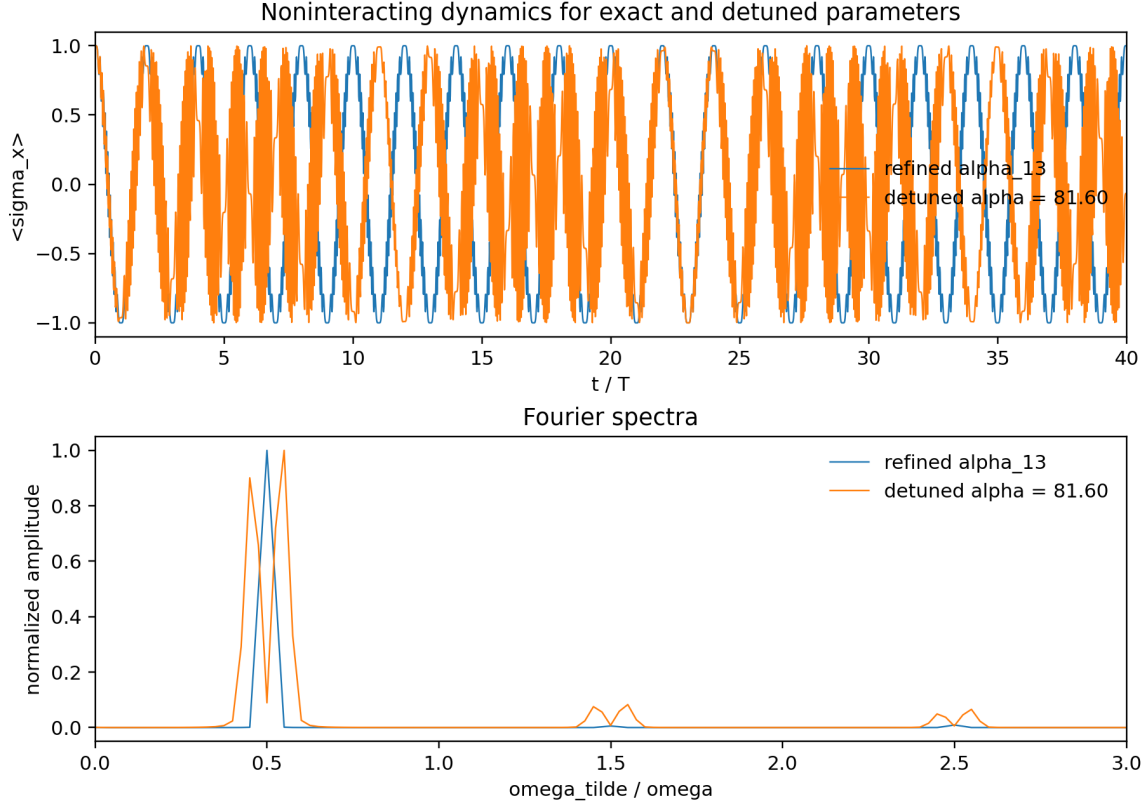


FIG. 1. Noninteracting $\sigma_x(t)$ comparison for refined α_{13} and detuned $\alpha = 81.60$.

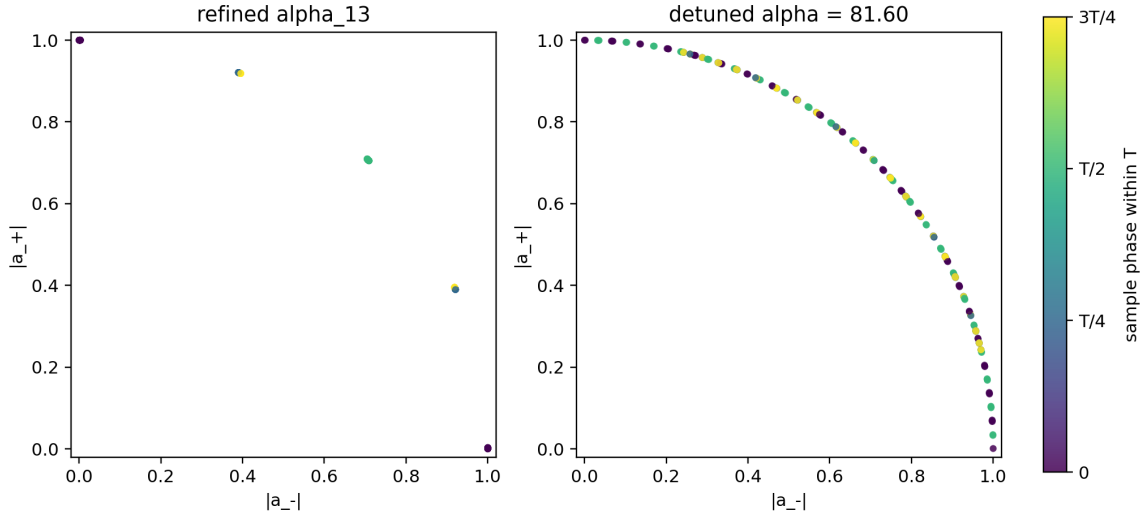


FIG. 2. Fractional-period projection comparison. Sampling within each period exposes the eigenstate exchange that is hidden in integer-period-only plots.

TABLE I. Noninteracting solver diagnostics. Frequencies are normalized by the drive angular frequency.

case	α	Magnus norm drift	RK4 norm drift	dominant frequency
refined α_{13}	80.0462421169	4.02×10^{-5}	1.59×10^{-3}	0.49984
detuned	81.60	3.92×10^{-5}	1.79×10^{-3}	0.54983

IV. INTERACTING REPRODUCTION

A. Implementation and boundary conditions

The interacting code evolves an exact state vector of dimension 2^L . The Hamiltonian action is implemented with bit-flip tensor indexing rather than building dense $2^L \times 2^L$ matrices for ordinary time traces. Long-time stroboscopic sampling uses a dense one-period Floquet operator and logarithmic sampling in n , which makes 10^{10} – 10^{12} period checks feasible for $L = 8$ and $L = 10$.

Open boundary conditions use $L - 1$ nearest-neighbor bonds. Periodic boundary conditions add the final $L, 1$ bond and are controlled by the JSON parameter `boundary`. Periodic outputs carry a suffix so that they do not overwrite open-boundary results.

The H100 package and local code use the same parameter structure. The current main settings are $L = 10$, open boundary conditions, $\alpha = 81.60$ and refined α_{13} , $J_{\text{paper}} = 0.2$, `interaction_operator_scale` = 0.5, and 120 steps per driving period for the one-period Floquet construction.

B. Short-time response

The first interacting check reproduces the half-drive-frequency response of $m_x(t)$ over the first 40 periods. Figure 4 shows the $L = 10$ time traces. The corresponding spectra have dominant normalized frequency 0.4998 for both the refined and detuned parameters over this short window.

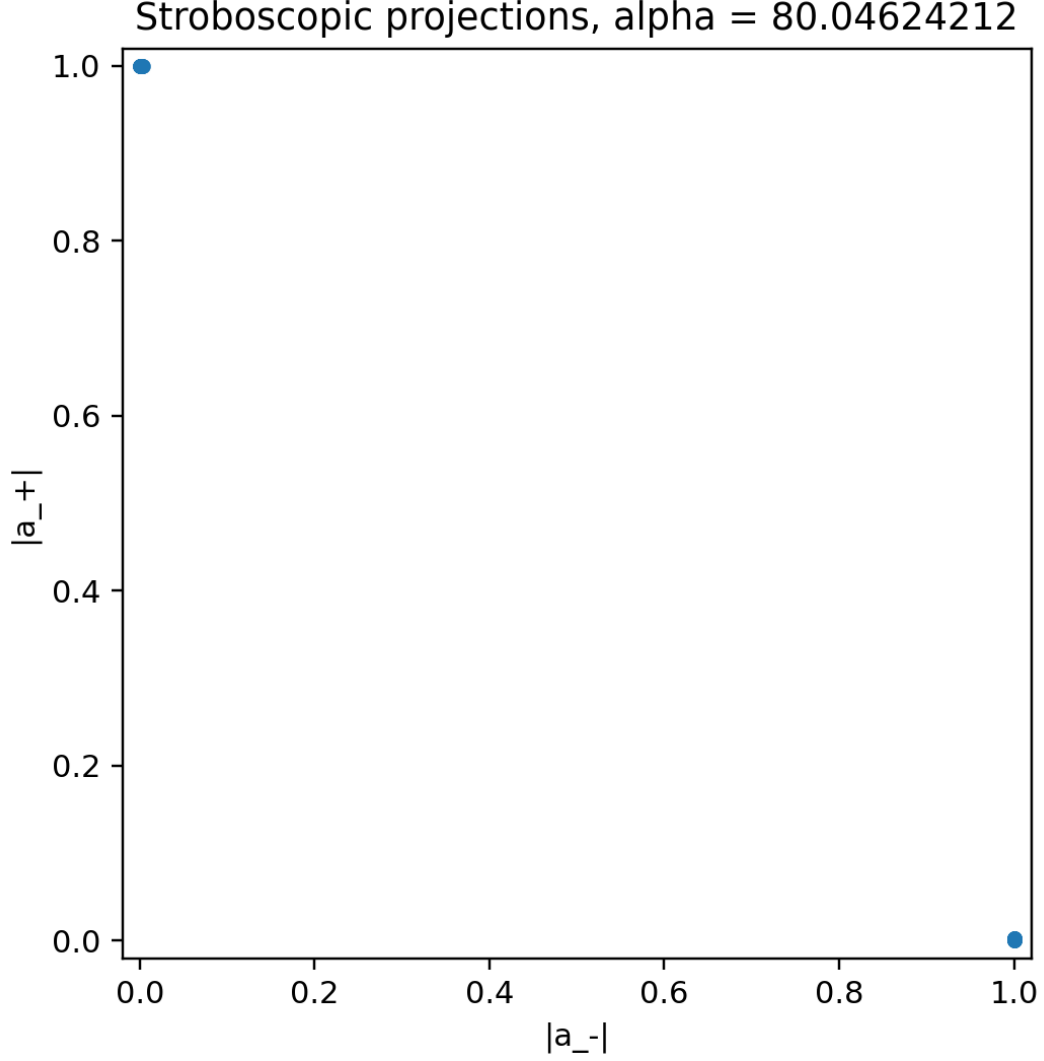


FIG. 3. Stroboscopic projection at refined α_{13} . The two-period structure follows from the nonsymmetric dynamical symmetry.

C. Interaction convention check

The most important debugging step was the comparison in Fig. 5. If the paper-label $J = 0.2$ is used directly as a Pauli-product coefficient, the $L = 10$, refined- α_{13} signal falls below $Z = 0.8$ by $n = 160$. This contradicts the expected stability and the paper-scale result. With the spin-operator convention $J_{\text{eff}} = 0.1$, the same $L = 10$ run stays above $Z = 0.9$ through the sampled 10^{10} periods.

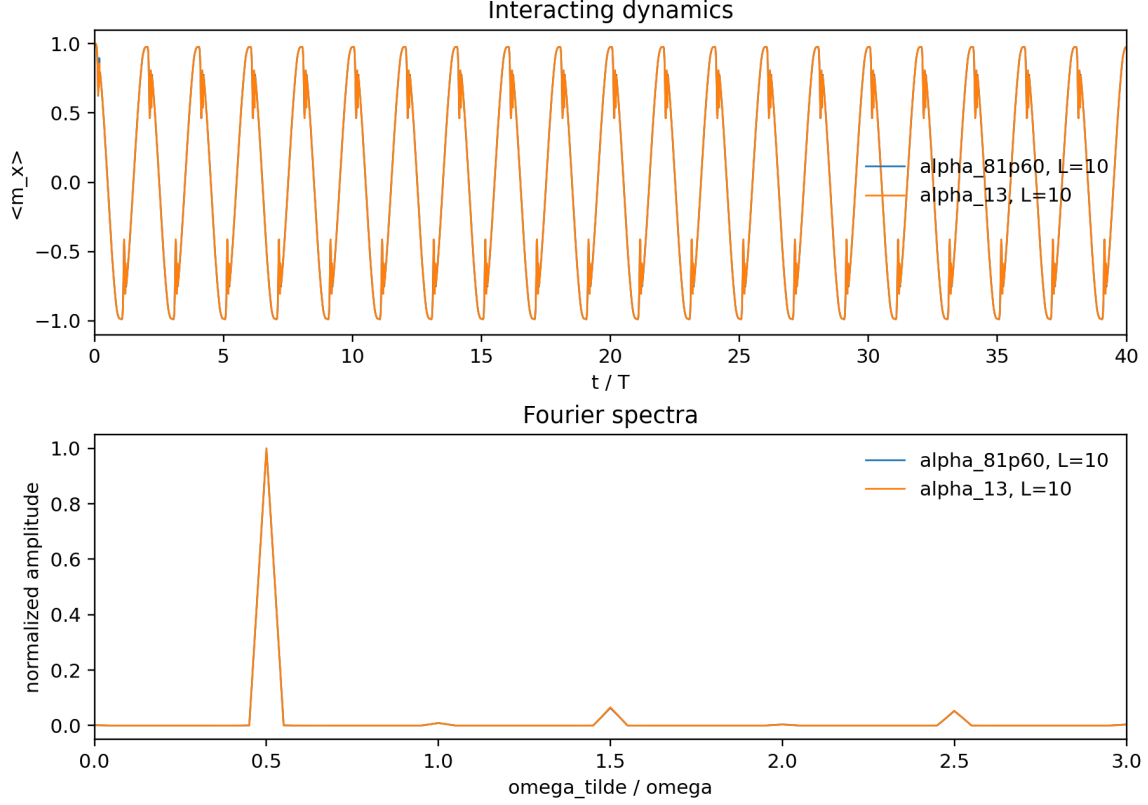


FIG. 4. Interacting $L = 10$ short-time $m_x(t)$ traces. Short-time period doubling is visible for both refined α_{13} and detuned $\alpha = 81.60$.

D. Finite-size comparison

The current-convention size comparison uses open boundaries, $J_{\text{eff}} = 0.1$, fourth-order Magnus one-period Floquet construction, double precision, and logarithmic stroboscopic sampling. The refined α_{13} case now shows the expected lifetime enhancement from $L = 8$ to $L = 10$, as shown in Fig. 6. Using the operational threshold $Z(n) < 0.8$, $L = 8$ crosses at $n = 1.9967 \times 10^8$, while $L = 10$ crosses at $n = 1.5094 \times 10^{10}$.

The detuned $\alpha = 81.60$ result remains the main discrepancy. Figure 7 shows that both $L = 8$ and $L = 10$ drop quickly in the raw $Z(n)$ trace. The threshold $Z(n) < 0.8$ is reached at $n = 3$ for both sizes, and the first negative sampled point occurs at $n = 65$ for $L = 8$ and $n = 78$ for $L = 10$. This is not consistent with a straightforward reading of the paper's Fig. 4(c).

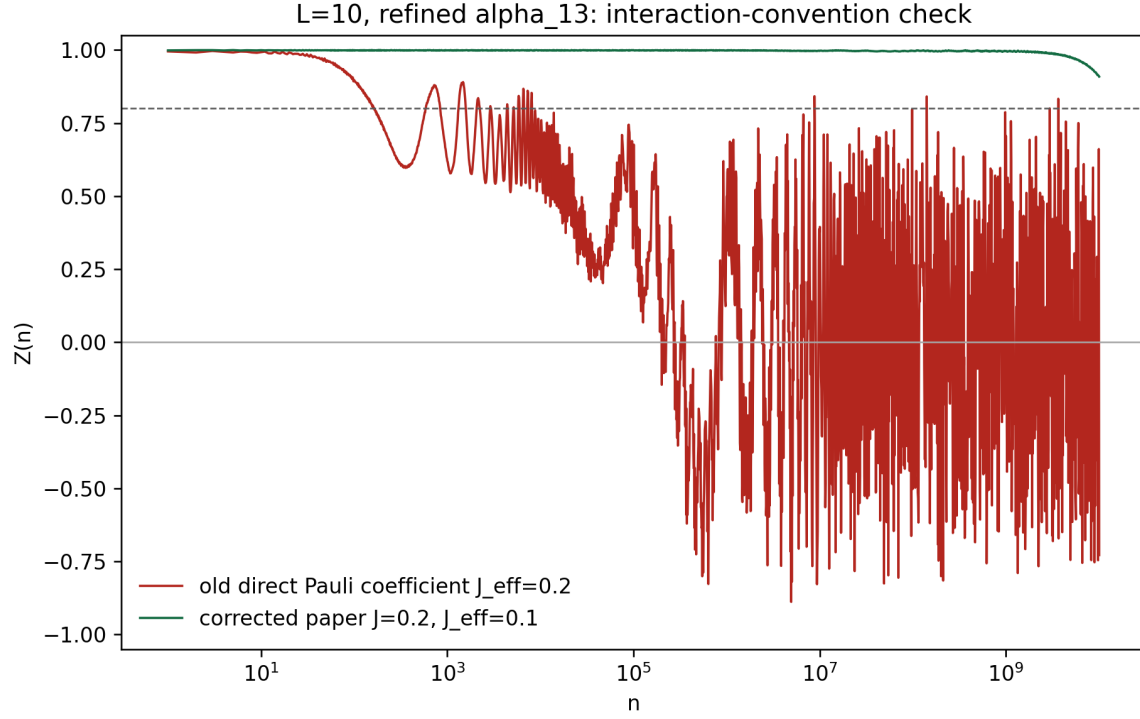


FIG. 5. Interaction-coefficient convention check for $L = 10$, refined α_{13} . The corrected convention maps the paper-label $J = 0.2$ to $J_{\text{eff}} = 0.1$ for Pauli products.

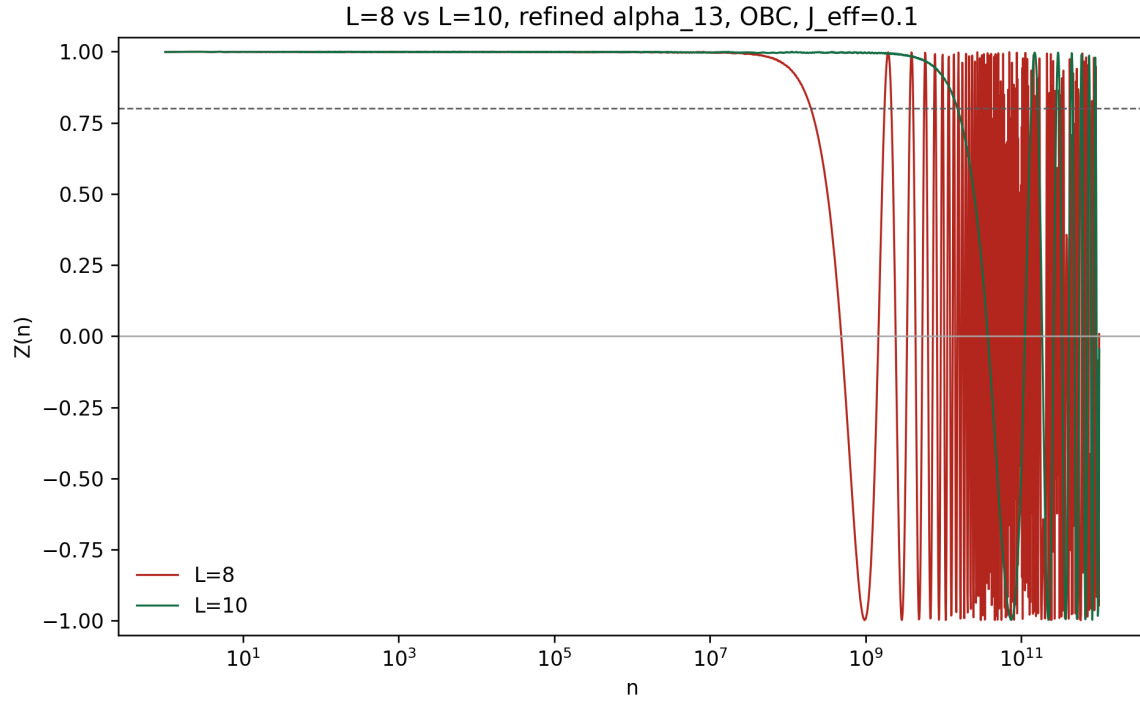


FIG. 6. Open-boundary $L = 8$ and $L = 10$ comparison for refined α_{13} using the corrected interaction convention. The larger chain has a much longer high- $Z(n)$ interval.

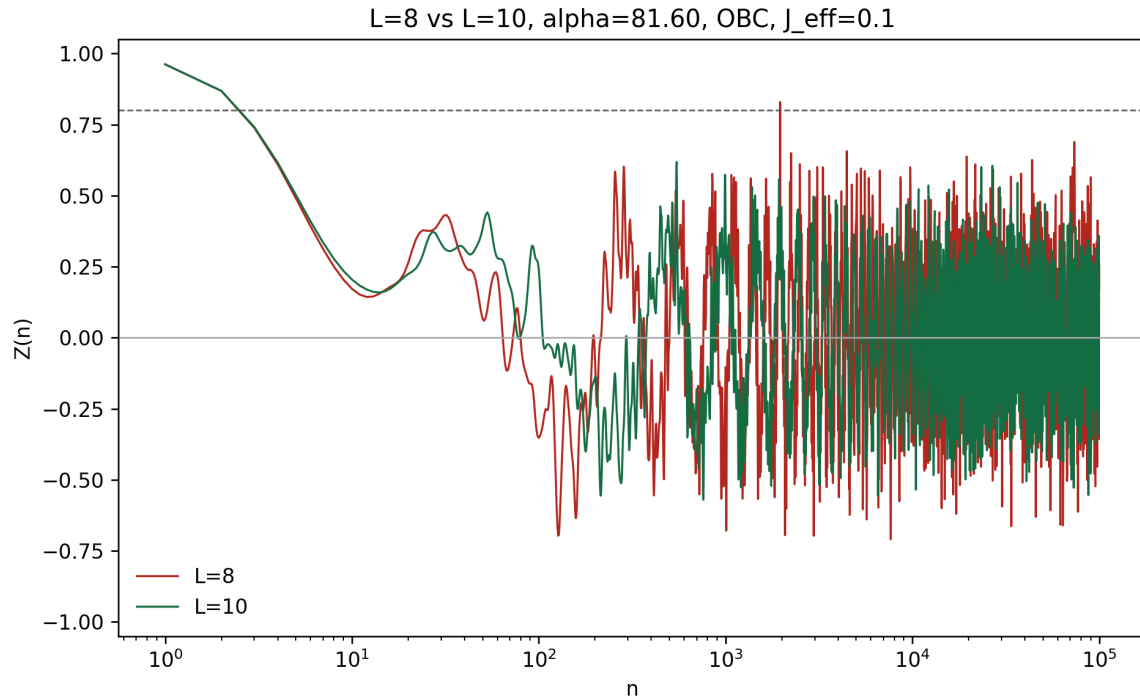


FIG. 7. Open-boundary $L = 8$ and $L = 10$ comparison for detuned $\alpha = 81.60$. The raw $Z(n)$ trace remains much less stable than expected from Fig. 4(c).

TABLE II. Current-convention open-boundary stroboscopic thresholds.

case	L	$n_{\max}$	first $Z < 0.8$	first $Z < 0$
$\alpha = 81.60$	8	10^5	3	65
$\alpha = 81.60$	10	10^5	3	78
refined α_{13}	8	10^{12}	1.9967×10^8	4.8793×10^8
refined α_{13}	10	10^{12}	1.5094×10^{10}	3.7263×10^{10}

E. Boundary-condition checks

The code originally used open boundaries. A periodic-boundary interface was added to test whether the detuned discrepancy was caused by boundary choice. For $L = 10$, refined α_{13} , periodic boundaries produced collapse and revival behavior, with first sampled $Z(n) < 0.8$ at $n = 7.8905 \times 10^8$ and first $Z(n) < 0$ at $n = 1.9224 \times 10^9$. For detuned $\alpha = 81.60$, both open and periodic boundaries still crossed $Z(n) < 0.8$ by $n = 3$. Boundary choice therefore does not explain the detuned Fig. 4(c) discrepancy.

V. REPRODUCIBILITY AND HPC WORKFLOW

The local scripts are:

- `reproduction_noninteracting/`
`reproduce_noninteracting.py`;
- `reproduction_interacting/`
`reproduce_interacting.py`.

The core interacting cluster launcher is `reproduction_interacting/run_interacting_hpc.sh`. It creates local `logs/` and `results/` directories before execution, records an environment snapshot, and excludes `gpuh01` because that node was known to have an older CUDA environment. The Slurm partition on the target cluster is `NV_H100`, not `gpu`.

The main local validation commands are

```
python reproduction_noninteracting\reproduce_noninteracting.py
python reproduction_interacting\reproduce_interacting.py ^
--params reproduction_interacting\params\interacting_params_hpc.json ^
--output-root reproduction_interacting --device cuda
```

and the report build command is

```
pdflatex -interaction=nonstopmode -halt-on-error comprehensive_report.tex
```

VI. ASSESSMENT

The reproduction is successful for the central symmetry-enforced mechanism and for the refined α_{13} interacting stability after the interaction convention is corrected. The following points are now well supported:

1. The noninteracting two-level model displays clean two-period dynamics at the refined α_{13} .
2. The fractional-period projections show the expected intra-period eigenstate exchange.
3. The interacting short-time $m_x(t)$ trace shows a half-drive-frequency peak.

4. The refined α_{13} interacting chain has a longer $Z(n)$ lifetime for $L = 10$ than for $L = 8$ under the corrected $J_{\text{eff}} = 0.1$ convention.

The main remaining risk is the detuned $\alpha = 81.60$ case. Under the current raw observable and both tested boundary conditions, the signal decays much faster than the paper’s Fig. 4(c). This could reflect a difference in lifetime definition, observable processing, interaction normalization, or the exact numerical protocol used for the paper. Before making a publication-level claim about the detuned finite-size scaling, the next checks should test alternative interaction normalizations, sign-corrected local autocorrelations, envelope-based lifetime extraction, and $L = 12$ on the H100 if memory allows.

VII. CONCLUSION

The project has now reproduced the main nonsymmorphic DTC mechanism and resolved the largest apparent contradiction in the refined- α_{13} interacting data. The final corrected results support the expected DTC picture: accurate-period dynamics is much more stable than detuned dynamics, and the refined α_{13} lifetime increases strongly from $L = 8$ to $L = 10$. The detuned Fig. 4(c) quantitative mismatch remains an explicit open problem rather than being hidden in the final report.

ACKNOWLEDGMENTS

This report summarizes local RTX 4090 calculations, H100-oriented package preparation, and locally ingested H100 output generated during the reproduction workflow.

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- [1] H. Hu, D. Fu, Y. Li, and S.-Q. Shen, “Solvable model for discrete time crystal enforced by nonsymmorphic dynamical symmetry,” *Phys. Rev. Research* **5**, L032024 (2023).
 - [2] D. V. Else, B. Bauer, and C. Nayak, “Floquet time crystals,” *Phys. Rev. Lett.* **117**, 090402 (2016).
 - [3] V. Khemani, A. Lazarides, R. Moessner, and S. L. Sondhi, “Phase structure of driven quantum systems,” *Phys. Rev. Lett.* **116**, 250401 (2016).

- [4] D. A. Abanin, W. De Roeck, W. W. Ho, and F. Huveneers, “Effective Hamiltonians, prethermalization, and slow energy absorption in periodically driven many-body systems,” *Phys. Rev. B* **95**, 014112 (2017).

SUPPLEMENTAL MATERIAL: EQUATIONS AND NUMERICAL METHODS

This supplemental section is a self-contained numerical guide. It is deliberately more explicit than the main report. The goal is that a reader who has seen undergraduate-level linear algebra, Taylor expansions, and ordinary differential equations can reconstruct the numerical methods used in the reproduction.

S1. Two-level Schrodinger equation as an ordinary differential equation

For the noninteracting model the state is a two-component spinor

$$|\psi(t)\rangle = \begin{pmatrix} \psi_{\uparrow}(t) \\ \psi_{\downarrow}(t) \end{pmatrix}. \quad (9)$$

The Schrodinger equation is

$$i\hbar \frac{d}{dt} |\psi(t)\rangle = H_0(t) |\psi(t)\rangle. \quad (10)$$

With $\hbar = 1$, this becomes the linear ordinary differential equation

$$\frac{d}{dt} |\psi(t)\rangle = A(t) |\psi(t)\rangle, \quad A(t) = -iH_0(t). \quad (11)$$

Using

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad (12)$$

Eq. (1) becomes

$$H_0(t) = \begin{pmatrix} 0 & a(t) - ib(t) \\ a(t) + ib(t) & 0 \end{pmatrix}, \quad (13)$$

where

$$a(t) = \frac{1}{2}\Omega \sin t, \quad b(t) = \Omega \sin^2 \left(\frac{t}{2} \right). \quad (14)$$

The matrix $H_0(t)$ is Hermitian, so $A(t) = -iH_0(t)$ is anti-Hermitian:

$$A^\dagger(t) = -A(t). \quad (15)$$

This one-line observation is important. The exact quantum propagator is unitary because it is generated by an anti-Hermitian matrix. A numerical method that preserves this anti-Hermitian exponential structure will preserve the norm much better than a generic polynomial time-stepper.

S2. Exact propagator and why time ordering matters

If $A(t)$ were constant over a time step, the exact solution would be

$$|\psi(t + \Delta t)\rangle = \exp[\Delta t A] |\psi(t)\rangle. \quad (16)$$

For time-dependent $A(t)$, the matrices at different times generally do not commute:

$$[A(t_1), A(t_2)] \neq 0. \quad (17)$$

The exact solution is therefore the time-ordered exponential

$$|\psi(t + \Delta t)\rangle = \mathcal{T} \exp \left[\int_t^{t+\Delta t} A(s) ds \right] |\psi(t)\rangle. \quad (18)$$

Here $\mathcal{T}$ means that later-time matrices must be ordered to the left of earlier-time matrices. If all matrices commute, time ordering is unnecessary and the solution reduces to an ordinary exponential. The noncommuting case is why both Magnus methods and high-order Runge–Kutta methods are useful.

S3. Instantaneous eigenstates and stroboscopic projections

At each time, $H_0(t)$ has two instantaneous eigenstates. The paper writes them in a rotating basis, where the nonsymmorphic dynamical symmetry makes the eigenstates exchange after one period. Numerically we do not need to symbolically solve the full Heun-function problem. Instead we evolve the state and project it onto the basis used by the paper:

$$P_\chi(t) = |\langle \phi_\chi(t) | \psi(t) \rangle|^2, \quad \chi = \pm. \quad (19)$$

The exact- α_n condition implies that an initial $\phi_+(0)$ state has almost zero probability to return to $\phi_+(0)$ after one period:

$$P_+(T) \simeq 0. \quad (20)$$

In the reproduction this condition is used to refine the tabulated α_{13} . The resulting local value is

$$\alpha_{13} = 80.04624211689759, \quad (21)$$

with one-period return probability 3.57×10^{-18} .

Integer-period projections show the period-doubled pattern, but they hide the motion inside each period. The reproduction therefore samples

$$t = nT, \quad nT + T/4, \quad nT + T/2, \quad nT + 3T/4. \quad (22)$$

This is why the final noninteracting data include both stroboscopic and fractional-period figures.

S4. Fourth-order Runge–Kutta from Taylor expansion

The classical fourth-order Runge–Kutta method is easiest to understand first for a general ordinary differential equation

$$\frac{dy}{dt} = f(t, y). \quad (23)$$

The exact solution over one step has the Taylor expansion

$$y(t + h) = y(t) + hy'(t) + \frac{h^2}{2}y''(t) + \frac{h^3}{6}y^{(3)}(t) + \frac{h^4}{24}y^{(4)}(t) + O(h^5). \quad (24)$$

The purpose of RK4 is to combine four slope evaluations so that the resulting formula matches Eq. (24) through the h^4 term.

The RK4 stages are

$$k_1 = f(t_n, y_n), \quad (25)$$

$$k_2 = f(t_n + h/2, y_n + hk_1/2), \quad (26)$$

$$k_3 = f(t_n + h/2, y_n + hk_2/2), \quad (27)$$

$$k_4 = f(t_n + h, y_n + hk_3). \quad (28)$$

The update is

$$y_{n+1} = y_n + h(b_1k_1 + b_2k_2 + b_3k_3 + b_4k_4). \quad (29)$$

The classical weights are

$$b_1 = \frac{1}{6}, \quad b_2 = \frac{1}{3}, \quad b_3 = \frac{1}{3}, \quad b_4 = \frac{1}{6}. \quad (30)$$

These coefficients are not arbitrary. Expanding each k_i in powers of h and matching Eq. (24)

gives the order conditions. The first few conditions are

$$b_1 + b_2 + b_3 + b_4 = 1, \quad (31)$$

$$b_2c_2 + b_3c_3 + b_4c_4 = \frac{1}{2}, \quad (32)$$

$$b_2c_2^2 + b_3c_3^2 + b_4c_4^2 = \frac{1}{3}, \quad (33)$$

$$b_4a_{43}a_{32}c_2 = \frac{1}{24}, \quad (34)$$

with

$$c_1 = 0, \quad c_2 = c_3 = \frac{1}{2}, \quad c_4 = 1, \quad a_{21} = a_{32} = \frac{1}{2}, \quad a_{43} = 1. \quad (35)$$

Solving the full set of fourth-order conditions gives the familiar RK4 tableau

$$\begin{array}{c|cccc} 0 & & & & \\ 1/2 & 1/2 & & & \\ 1/2 & 0 & 1/2 & & \\ 1 & 0 & 0 & 1 & \\ \hline & 1/6 & 1/3 & 1/3 & 1/6 \end{array}. \quad (36)$$

For the Schrodinger equation, y is the state vector ψ , and

$$f(t, \psi) = -iH(t)\psi. \quad (37)$$

Nothing in the RK4 derivation requires y to be a scalar. The same formula applies when y is a vector and f contains a matrix-vector product. In implementation, each k_i is itself a vector of the same dimension as ψ .

S5. RK4 accuracy and loss of unitarity

Because the RK4 formula matches the exact Taylor expansion through h^4 , the local truncation error per step is

$$\epsilon_{\text{local}} = O(h^5). \quad (38)$$

For a fixed final time T_{final} , the number of steps is approximately $N = T_{\text{final}}/h$, so accumulated global error scales as

$$\epsilon_{\text{global}} = NO(h^5) = O(h^4). \quad (39)$$

This is the sense in which RK4 is a fourth-order method.

The important limitation is that RK4 is a polynomial approximation to the exponential propagator, not an exponential itself. For a constant Hamiltonian, $A = -iH$, one RK4 step gives

$$\psi_{n+1} = R(hA)\psi_n, \quad (40)$$

where

$$R(z) = 1 + z + \frac{z^2}{2} + \frac{z^3}{6} + \frac{z^4}{24}. \quad (41)$$

This is the degree-four Taylor polynomial of e^z . If A is anti-Hermitian, the exact e^{hA} is unitary. However, $R(hA)$ is generally not unitary. For a single eigenmode with $z = ix$,

$$|R(ix)|^2 = 1 - \frac{x^6}{72} + O(x^8), \quad (42)$$

so even for a constant Hermitian Hamiltonian the RK4 step can slowly change the norm. The error is high order and usually small for small h , but it is not exactly zero.

This explains the diagnostics in the reproduction. RK4 is useful as an independent high-order check, but the norm drift must be monitored:

$$\delta_{\text{norm}} = \max_t |\langle \psi(t) | \psi(t) \rangle - 1|. \quad (43)$$

In the noninteracting checks the RK4 drift is about 10^{-3} , while the Magnus drift is about 10^{-5} for the chosen time grid. For the interacting time-trace code, optional normalization after each RK4 step is used only as a practical stabilizer; it does not make RK4 a genuinely unitary integrator.

S6. Magnus expansion and its logic

The Magnus idea is to keep the exact exponential structure:

$$\psi(t+h) = \exp[\Omega(t,h)]\psi(t). \quad (44)$$

The generator Ω is chosen so that this exponential reproduces the time-ordered exponential. The first terms of the Magnus expansion are

$$\Omega_1 = \int_t^{t+h} A(t_1) dt_1, \quad (45)$$

$$\Omega_2 = \frac{1}{2} \int_t^{t+h} dt_1 \int_t^{t_1} dt_2 [A(t_1), A(t_2)], \quad (46)$$

$$\Omega_3 = \frac{1}{6} \int_t^{t+h} dt_1 \int_t^{t_1} dt_2 \int_t^{t_2} dt_3 \left([A_1, [A_2, A_3]] + [A_3, [A_2, A_1]] \right), \quad (47)$$

where $A_j = A(t_j)$. The commutators appear because matrix multiplication is noncommutative. If all $A(t)$ commute, every commutator vanishes and the Magnus expansion reduces to the simple integral $\Omega = \int A(t) dt$.

For Schrodinger evolution $A(t) = -iH(t)$ is anti-Hermitian. The commutator of two anti-Hermitian matrices is also anti-Hermitian:

$$\begin{aligned}
[A, B]^\dagger &= (AB - BA)^\dagger \\
&= B^\dagger A^\dagger - A^\dagger B^\dagger \\
&= (-B)(-A) - (-A)(-B) \\
&= BA - AB \\
&= -[A, B].
\end{aligned} \tag{48}$$

Therefore each retained Magnus term is anti-Hermitian. If the numerical approximation Ω_{num} is anti-Hermitian, then

$$U_{\text{num}} = \exp(\Omega_{\text{num}}) \tag{49}$$

is unitary:

$$U_{\text{num}}^\dagger U_{\text{num}} = \exp(\Omega_{\text{num}}^\dagger) \exp(\Omega_{\text{num}}) = \exp(-\Omega_{\text{num}}) \exp(\Omega_{\text{num}}) = I. \tag{50}$$

This is the structural reason the Magnus method is better suited to quantum evolution than a generic Taylor polynomial method.

S7. Fourth-order Gauss–Legendre Magnus step

The reproduction uses a standard fourth-order two-point Gauss–Legendre Magnus step. Let

$$t_1 = t + \left(\frac{1}{2} - \frac{\sqrt{3}}{6}\right) h, \quad t_2 = t + \left(\frac{1}{2} + \frac{\sqrt{3}}{6}\right) h, \tag{51}$$

and define

$$A_1 = A(t_1), \quad A_2 = A(t_2). \tag{52}$$

Two-point Gauss–Legendre quadrature integrates ordinary scalar functions exactly through cubic order. Applying the same idea to the matrix-valued integral gives

$$\Omega_1 = \int_t^{t+h} A(s) ds = \frac{h}{2}(A_1 + A_2) + O(h^5). \tag{53}$$

The leading noncommutative correction comes from Ω_2 . Keeping terms through fourth order yields

$$\Omega_{\text{GL4}} = \frac{h}{2}(A_1 + A_2) - \frac{\sqrt{3}h^2}{12}[A_1, A_2]. \quad (54)$$

The sign convention follows the ordering used in Eq. (11) and in the code. Some textbooks write the same formula with the commutator order reversed, which changes the sign because $[A_2, A_1] = -[A_1, A_2]$.

The numerical update is

$$\psi_{n+1} = \exp(\Omega_{\text{GL4}})\psi_n. \quad (55)$$

For a smooth $A(t)$, this has local error

$$\epsilon_{\text{local}} = O(h^5), \quad (56)$$

and global error

$$\epsilon_{\text{global}} = O(h^4). \quad (57)$$

Thus it is fourth order, like RK4, but it also preserves the unitary structure of Schrodinger evolution at the level of each step.

In the two-level calculation, Ω_{GL4} is a 2×2 matrix and the matrix exponential is cheap. In the interacting Floquet construction, Ω_{GL4} is a $2^L \times 2^L$ matrix for the dense one-period calculation. The step matrices are multiplied to build

$$U(T) \approx U_{N-1} \cdots U_1 U_0, \quad U_j = \exp(\Omega_{\text{GL4},j}). \quad (58)$$

Since each U_j is unitary up to floating-point matrix-exponential error, the product is also unitary up to roundoff.

S8. Comparing RK4 and Magnus in this reproduction

Both methods are fourth order, but they control different numerical risks:

- RK4 is simple, explicit, and works for general nonlinear ODEs. It provides an independent benchmark for short-time traces. Its main weakness here is that it does not exactly preserve quantum unitarity.

- Fourth-order Magnus is specialized for linear matrix evolution. It is more expensive because of matrix exponentials and commutators, but it respects the exponential and unitary structure of quantum mechanics.

The reproduction uses this division of labor. RK4 checks whether the waveform is stable under an independent integrator. Magnus is used for unitary-sensitive evolution and for one-period Floquet operators. Agreement between the two over short times gives confidence that the observed period doubling is not an artifact of one particular numerical method.

S9. Many-body basis and Hamiltonian action

For the interacting chain the computational basis is the z -basis

$$|b_{L-1} \cdots b_1 b_0\rangle, \quad b_i \in \{0, 1\}. \quad (59)$$

The Hilbert-space dimension is 2^L . A basis index is stored as an integer

$$q = \sum_{i=0}^{L-1} b_i 2^i. \quad (60)$$

Acting with σ_x^i flips bit i :

$$q \mapsto q \oplus 2^i, \quad (61)$$

where $\oplus$ is bitwise exclusive-or. Acting with σ_y^i also flips the bit but adds a phase:

$$\sigma_y|0\rangle = i|1\rangle, \quad \sigma_y|1\rangle = -i|0\rangle. \quad (62)$$

This lets the code apply the Hamiltonian to the state vector without building a dense many-body Hamiltonian for ordinary time traces.

The interaction term is diagonal in the x -flipped representation:

$$\sigma_x^i \sigma_x^{i+1} |q\rangle = |q \oplus 2^i \oplus 2^{i+1}\rangle. \quad (63)$$

For open boundaries the bond list is

$$(1, 2), (2, 3), \dots, (L-1, L). \quad (64)$$

For periodic boundaries the additional bond $(L, 1)$ is included.

S10. Initial state and observable evaluation

The all- $+x$ product state is represented in the z -basis as

$$|+x\rangle^{\otimes L} = 2^{-L/2} \sum_q |q\rangle. \quad (65)$$

If a site is initialized in $|-x\rangle$, the code changes the sign of amplitudes with $b_i = 1$. The current production runs use the all- $+x$ case because it directly maximizes $m_x(0)$ and matches the main-text Fig. 4 convention.

For a normalized state vector ψ_q , the local x -magnetization is

$$\langle \sigma_x^i \rangle = \sum_q \psi_q^* \psi_{q \oplus 2^i}. \quad (66)$$

The averaged magnetization and DTC diagnostic are

$$m_x(nT) = \frac{1}{L} \sum_i \langle \sigma_x^i(nT) \rangle, \quad Z(n) = (-1)^n m_x(nT). \quad (67)$$

S11. Long-time Floquet sampling

Directly evolving every period up to $n = 10^{10}$ or 10^{12} is impractical. The reproduction instead constructs the one-period Floquet matrix $U(T)$ and uses logarithmic sample points. For each sample n , the state is computed by matrix-power action:

$$|\psi(nT)\rangle = U(T)^n |\psi(0)\rangle. \quad (68)$$

The sample set contains a dense linear prefix for early-time behavior and logarithmically spaced points for long-time decay. This keeps the CSV files small while preserving the lifetime-scale information.

For $L = 10$, $U(T)$ is a 1024×1024 complex matrix, which is feasible on a local RTX 4090 and on the H100 node. For $L = 12$, the dimension becomes 4096, so memory and runtime must be checked before launching long sweeps.

S12. Practical reproduction checklist

A minimal local reproduction proceeds as follows:

1. Set $\hbar = 1$, $\omega = 1$, $T = 2\pi$, and $\Omega = \alpha/8$.

2. For the noninteracting model, evolve Eq. (11) using both Eq. (54) and RK4.
3. Refine α_{13} by minimizing the one-period return probability.
4. For the interacting model, initialize $|+x\rangle^{\otimes L}$.
5. Use the paper-label $J = 0.2$ but apply the Pauli-product coefficient $J_{\text{eff}} = 0.1$.
6. Build $U(T)$ with fourth-order Magnus steps and sample $Z(n)$ at linear-plus-logarithmic periods.
7. Compare $Z(n)$ thresholds across L , boundary conditions, and α .

The final data show that refined α_{13} passes this checklist and has the expected longer lifetime at larger L . The detuned $\alpha = 81.60$ case remains the only substantial quantitative mismatch with the paper's plotted Fig. 4(c).